Specification Shifts and Included Variable Bias

A formal note separating projection algebra from causal interpretation

Manoel Galdino, Davi Moreira, and Carolina Dolleans

11 July 2026

Contents

1 Purpose and scope

This note separates three objects that must not be conflated:

1. a population shift between two nested linear projections;
2. its exact finite-sample analogue; and
3. bias relative to a causal target, here denoted by the contemporaneous treatment effect (CET).

The first two objects are defined by projection algebra. They do not require a causal model. The third requires a target estimand, a time ordering, a causal graph, and identification assumptions. Accordingly, this note uses *specification shift* as the general term and reserves *included variable bias* for cases in which those additional causal conditions establish that including Z moves an estimator away from the CET.

The derivation is scalar: D is the focal regressor and Z is one candidate control. The vector case, uncertainty quantification, and identification of the CET are separate tasks.

2 Setup: two nested linear projections

Let Y, D, and Z be scalar random variables with finite second moments. Let W be a column vector of common regressors that includes an intercept. In a TSCS application, W may also include unit and time effects, lagged outcomes, lagged treatments, and other predetermined state variables. Nothing in the algebra below establishes that the variables in W are causally admissible.

Define the short population linear projection by

$$(\beta_S, \gamma_S) = \arg\min_{b,g} \mathbb{E}\left[(Y - bD - W'g)^2\right], \tag{1}$$

and the long population linear projection by

$$(\beta_L, \theta_L, \gamma_L) = \arg\min_{b,t,g} \mathbb{E}\left[(Y - bD - tZ - W'g)^2\right]. \tag{2}$$

The labels short and long describe regressor sets only. Neither label means true, correct, unbiased, or causal.

Let $P_W A$ denote the population linear projection of a scalar random variable A on W, and write

$$\widetilde{A}_W = A - P_W A.$$

Assume

$$0 < \mathbb{E}[\widetilde{D}_W^2] < \infty$$

and that the relevant population second-moment matrices have full rank. Define the auxiliary population projection

$$Z = \pi D + W'\kappa + r, \qquad \mathbb{E}[Dr] = 0, \qquad \mathbb{E}[Wr] = 0. \tag{3}$$

By partial regression,

$$\pi = \frac{\mathbb{E}[\widetilde{D}_W \widetilde{Z}_W]}{\mathbb{E}[\widetilde{D}_W^2]}. \tag{4}$$

3 Three distinct objects

3.1 Population specification shift

The population specification shift associated with adding Z, holding W and the population distribution fixed, is

$$\Delta_Z := \beta_L - \beta_S. \tag{5}$$

This is a contrast between two coefficients of linear projection. It exists without potential outcomes, a DAG, or an identification claim. Its value depends on the distribution of the variables and on the common regressor set W.

3.2 Sample specification shift

For a fixed sample, let $\hat{\beta}_S$ and $\hat{\beta}_L$ be the OLS coefficients on D in the short and long regressions, estimated using exactly the same observations and the same W. Define

$$\widehat{\Delta}_Z := \hat{\beta}_L - \hat{\beta}_S. \tag{6}$$

This is an observed coefficient movement. It is not, by definition, an estimate of causal bias.

3.3 Bias relative to the CET

Let β_{CET} be a scalar causal estimand fixed before comparing the two regressions. This note does not define the substantive timing or assumptions that identify it. For a population projection coefficient β_m, where $m \in \{S, L\}$, define its discrepancy from the causal target as

$$B_m^P := \beta_m - \beta_{\text{CET}}. \tag{7}$$

For a finite-sample estimator, the conventional bias is

$$\text{Bias}(\hat{\beta}_m; \beta_{\text{CET}}) := \mathbb{E}[\hat{\beta}_m] - \beta_{\text{CET}}, \tag{8}$$

provided the expectation exists and the sampling scheme is specified. These quantities are meaningful as *causal* discrepancies only after the CET and the identifying assumptions have been stated.

Because the same target is subtracted from both population coefficients,

$$\Delta_Z = B_L^P - B_S^P. \tag{9}$$

Thus, a specification shift is the difference between two target discrepancies. It does not reveal either discrepancy separately. Likewise, when both estimators use the same sampling scheme,

$$\mathbb{E}[\widehat{\Delta}_Z] = \text{Bias}(\hat{\beta}_L; \beta_{\text{CET}}) - \text{Bias}(\hat{\beta}_S; \beta_{\text{CET}}). \tag{10}$$

Equation (10) still does not establish that either estimator identifies the CET.

4 Population FWL identity

Proposition 1 (Scalar population specification-shift identity)**.** *Under the moment and rank conditions above,*

$$\boxed{\Delta_Z = \beta_L - \beta_S = -\theta_L \pi}. \tag{11}$$

This is an identity between coefficients of nested population linear projections.

Proof. Residualize Y, D, and Z with respect to W. The normal equation for D in the residualized long projection is

$$\mathbb{E}[\widetilde{D}_W \widetilde{Y}_W] = \beta_L \mathbb{E}[\widetilde{D}_W^2] + \theta_L \mathbb{E}[\widetilde{D}_W \widetilde{Z}_W].$$

Therefore,

$$\beta_L = \frac{\mathbb{E}[\widetilde{D}_W \widetilde{Y}_W]}{\mathbb{E}[\widetilde{D}_W^2]} - \theta_L \frac{\mathbb{E}[\widetilde{D}_W \widetilde{Z}_W]}{\mathbb{E}[\widetilde{D}_W^2]}.$$

FWL applied to the short projection gives

$$\beta_S = \frac{\mathbb{E}[\widetilde{D}_W \widetilde{Y}_W]}{\mathbb{E}[\widetilde{D}_W^2]},$$

and Equation (4) gives the second ratio as π. Subtraction yields

$$\beta_L - \beta_S = -\theta_L \pi.$$

$\square$

4.1 Equivalent projection-substitution check

The same result follows without using the coefficient formula directly. Write the long projection as

$$Y = \beta_L D + \theta_L Z + W' \gamma_L + u_L,$$

where u_L is orthogonal to D, Z, and W. Substituting Equation (3) gives

$$Y = (\beta_L + \theta_L \pi) D + W'(\gamma_L + \theta_L \kappa) + (u_L + \theta_L r).$$

The final composite residual is orthogonal to D and W. Uniqueness of the short projection therefore implies

$$\beta_S = \beta_L + \theta_L \pi,$$

which is equivalent to Equation (11). This second derivation independently verifies the sign.

5 Exact sample identity

Let y, d, and z be $n \times 1$ sample vectors and let W be the common $n \times k$ design matrix. Define the residual-maker

$$M_W = I - W(W'W)^{-1}W'.$$

Assume W has full column rank, $d' M_W d > 0$, and the augmented design matrix has full column rank. Then

$$\hat{\beta}_S = \frac{d' M_W y}{d' M_W d}$$

and the normal equation for D in the long regression gives

$$d' M_W y = \hat{\beta}_L d' M_W d + \hat{\theta}_L d' M_W z.$$

The coefficient on D in the sample auxiliary regression of Z on D and W is

$$\hat{\pi} = \frac{d' M_W z}{d' M_W d}.$$

Combining the three expressions yields the finite-sample identity

$$\boxed{\widehat{\Delta}_Z = \hat{\beta}_L - \hat{\beta}_S = -\hat{\theta}_L \hat{\pi}}. \tag{12}$$

No probability limit, exogeneity condition, homoskedasticity assumption, or causal claim is needed for Equation (12). Clustered standard errors also do not affect this point-estimate identity. However, exact equality requires the short, long, and auxiliary regressions to use the same observations, the same W, and the same estimation weights. If Z has missing values, the short model must be re-estimated on the long-model sample before applying the identity.

In a fixed-effects or ADL regression, fixed effects and the common dynamic regressors are components of W. The identity survives their residualization. This fact does not remove dynamic-panel bias, invalid controls in W, misspecified dynamics, or any other source of discrepancy from the CET.

6 Causal classification of the same algebra

The identity does not classify Z. A time-indexed DAG and substantive ordering must do that work. Table 1 states the strongest warranted interpretation in four leading cases. Each statement is conditional, not automatic.

6.1 Collider interpretation

A collider label is not enough by itself. To call the shift *included variable bias*, one must also establish that the short adjustment set identifies the CET and that adding Z opens a path that invalidates the long adjustment set. Under those conditions,

$$\beta_S = \beta_{\text{CET}} \quad \Longrightarrow \quad \Delta_Z = \beta_L - \beta_{\text{CET}}.$$

If the short model is already confounded or misspecified, the same Δ_Z is not the total bias of the long model.

6.2 Mediator interpretation with a total-effect target

If $D \to Z \to Y$ and the target is the total CET, conditioning on Z can remove part of the causal pathway. But the regression coefficient on D in the long model is not automatically a direct causal effect. That interpretation requires stronger mediation conditions than the FWL identity: well-defined interventions, appropriate control of mediator-outcome confounding, no treatment-induced mediator-outcome confounding for standard identification strategies, and functional-form restrictions when using a single linear coefficient.

If the short model identifies the total CET, the shift is the long projection's discrepancy from that target. Only in a compatible linear additive mediation model, without relevant interactions, can the shift be interpreted as the negative of an indirect component.

6.3 Confounder interpretation

If Z is a genuine pre-treatment common cause and the long adjustment set identifies the CET, then

$$\beta_L = \beta_{\text{CET}} \quad \Longrightarrow \quad \Delta_Z = -(\beta_S - \beta_{\text{CET}}).$$

The same algebra that records harmful movement in the collider case records corrective movement here. The sign of $-\theta_L \pi$ cannot distinguish these cases.

Table 1: Causal interpretations of the scalar specification shift, conditional on additional assumptions

Role of Z	Conditions needed for a causal comparison	Interpretation of $\Delta_Z = \beta_L - \beta_S$ under those conditions	What the identity alone cannot show
Collider	W suffices to identify the CET in the short specification; Z is a conditioned common effect or descendant that opens an otherwise blocked noncausal path; functional-form conditions map the short coefficient to the CET.	Then $B_S^P = 0$, so $\Delta_Z = B_L^P$. In this restricted case, the population shift is the discrepancy introduced by conditioning on Z.	That Z is a collider; that the short model identifies the CET; or that all of the shift is collider bias when other misspecification remains.
Mediator, total-effect target	The CET is explicitly the total contemporaneous effect; the short specification identifies that target. Interpreting the long coefficient as a direct effect additionally requires a suitable mediation model, including assumptions about mediator-outcome confounding and interactions.	If the short coefficient identifies the total CET, then $B_S^P = 0$ and $\Delta_Z = B_L^P$ relative to that total-effect target. Under stronger linear mediation conditions, the shift may equal the negative of an indirect component.	That the long coefficient generally identifies a controlled or natural direct effect, or that the product is a mediated effect without the additional mediation assumptions.
Confounder	Z is measured before treatment for the chosen CET; W and Z jointly satisfy the needed exchangeability and functional-form conditions; the long specification identifies the CET.	Then $B_L^P = 0$, so $\Delta_Z = -B_S^P$. The movement is a correction of the short projection's discrepancy under these assumptions.	That including Z is sufficient for identification; that measurement, overlap, timing, or model-form problems are absent; or that any nonzero shift is deconfounding.
Ambiguous or dual role	Z plausibly blocks one open path but opens or blocks another, or its within-period ordering is not known. No single identifying model is yet established.	$\Delta_Z = B_L^P - B_S^P$ is only the net difference between two possibly nonzero discrepancies. It may combine deconfounding, collider selection, over-control, and model-form changes.	Which component dominates; which coefficient is closer to the CET; or whether either coefficient has a causal interpretation.

6.4 Ambiguous and dual-role interpretation

A variable may be a confounder through one path and a collider or mediator through another. Alternatively, a substantive variable may be a confounder at $t-1$ but a collider or mediator at t. In either situation, neither the short nor the long coefficient is guaranteed to identify the CET.

For a same-index dual-role Z, the scalar shift is a net projection contrast. It does not separately recover the correction from closing a confounding path and the distortion from opening a collider path or blocking a mediating path. For a role that changes across time, Z_{t-1} and Z_t are distinct candidate regressors. The scalar identity applies to adding one candidate while holding the rest of W fixed; it does not by itself justify replacing Z_t with Z_{t-1}.

Therefore, ambiguity is a substantive finding, not a defect in the diagnostic. In that case the defensible report is the specification shift plus the competing causal interpretations, not a claim that the shift is bias.

7 Mathematical checks

The following checks validate the derivation without simulated data.

1. **Normal-equation check.** The FWL proof derives the identity directly from the long and short normal equations after residualizing by the same W.
2. **Independent substitution check.** Substituting the auxiliary projection of Z into the long projection gives $\beta_S = \beta_L + \theta_L \pi$, confirming both the product and its negative sign in Δ_Z.
3. **Finite-sample matrix check.** Equation (12) follows from exact OLS normal equations using the residual-maker M_W. No asymptotic approximation is used.
4. **Units check.** θ_L has units Y/Z and π has units Z/D. Their product has units Y/D, matching $\beta_L - \beta_S$.
5. **Rescaling check.** Replacing Z by cZ, with $c \neq 0$, changes θ_L to θ_L/c and π to $c\pi$. The product and the specification shift are invariant.
6. **Zero-component check.** If $\theta_L = 0$ or $\pi = 0$, then $\Delta_Z = 0$. This says only that the treatment coefficient does not move in the chosen linear projections; it does not establish causal validity or the absence of other bias.
7. **Target-cancellation check.** Subtracting the same β_{CET} from both population coefficients yields Equation (9). Hence the specification shift is a difference in discrepancies, not a discrepancy level.

8 Terminology rule

The following terminology keeps the algebra and the causal claim separate:

- **Population specification shift:** $\Delta_Z = \beta_L - \beta_S = -\theta_L \pi$.
- **Sample specification shift:** $\widehat{\Delta}_Z = \widehat{\beta}_L - \widehat{\beta}_S = -\widehat{\theta}_L \widehat{\pi}$, computed on a common sample.
- **Included variable bias:** use only when a stated causal graph, timing, target, and identification assumptions establish that the long specification is farther from β_{CET} because it conditions on Z.
- **Ambiguous shift:** use when the causal role is dual, contested, or unidentified.

A large absolute shift is evidence of specification sensitivity. It is not evidence, by itself, that Z should be included or excluded.

9 Unresolved issues and boundaries

This note intentionally leaves the following issues unresolved:

1. The formal definition and identification conditions for β_{CET}, including within-period timing, treatment histories, anticipation, and interference.
2. Finite-T bias in dynamic fixed-effects estimators. This is a distinct estimator problem and is not absorbed by the FWL identity.
3. Sampling uncertainty for Δ_Z, including covariance between nested estimators and cluster dependence.
4. Multiple correlated candidate controls, order dependence, and joint versus leave-one-out shifts.
5. Nonlinear links, non-nested specifications, regularized estimation, and comparisons that use different samples or different weights.
6. Identification of separate confounding, collider, and mediation components for a dual-role variable.

These boundaries are substantive. None can be resolved by the scalar product identity alone.